

Lesson 7

Installing and Configuring an OS

Topic 7A

Install an Operating System

Exam Objectives Covered

- 2.1 Given a scenario, install server operating systems

Installation Process

- OS source files
- Live installations
- The installer
- Manual versus unattended

```
[root@localhost ~]# cat anaconda-ks.cfg
# Generated by Anaconda 33.25.4
# Generated by pykickstart v3.29
#version=F33
# Use graphical install
graphical

# Keyboard layouts
keyboard --xlayouts='us'
# System language
lang en_US.UTF-8

# Network information
network --hostname=localhost.localdomain

# Run the Setup Agent on first boot
firstboot --enable

# Generated using Blivet version 3.3.0
ignoredisk --only-use=sda
autopart
# Partition clearing information
clearpart --none --initlabel

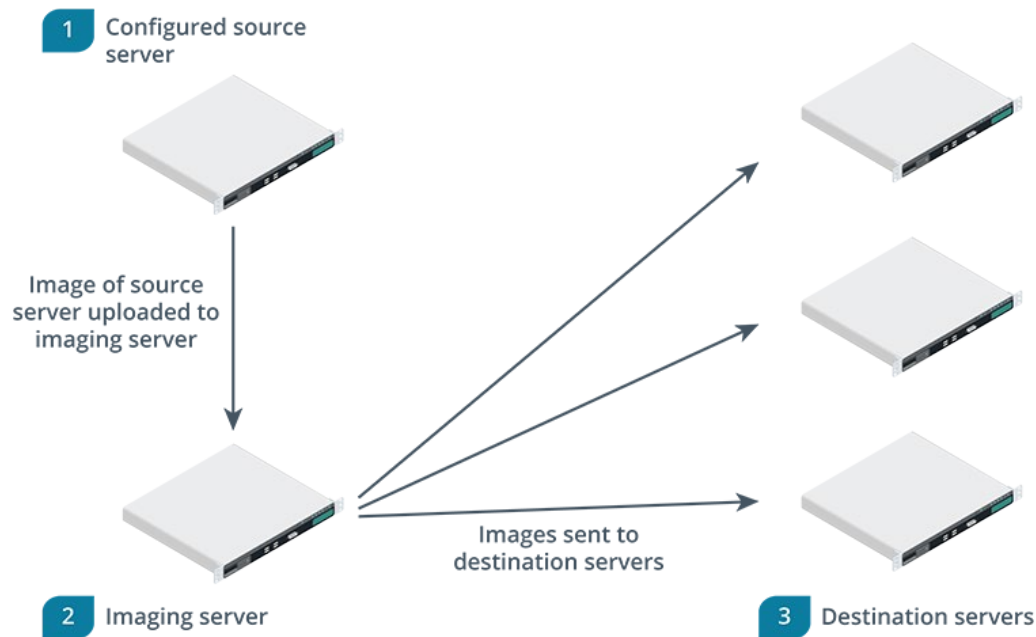
# System timezone
timezone America/Denver --utc

#Root password
rootpw --lock
```

Linux Kickstart files are used for automated, unattended installations.

Installation Process (slide 2)

- Imaging



The source computer's configuration is uploaded to a deployment server. The image is then deployed to a destination server. (Images © 123RF.com)

Installation Process (slide 3)

- Server migrations
- Bare metal versus virtualized installations

Managing the Server with a GUI or CLI

- Linux

```
[root@localhost ~]# uptime
07:58:31 up 54 min,  1 user,  load average: 0.00, 0.07, 0.04
[root@localhost ~]# hostname
localhost.localdomain
[root@localhost ~]# ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:12:67:76 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83120sec preferred_lft 83120sec
    inet6 fe80::4ba0:590f:a7ca:1827/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@localhost ~]# █
```

Linux administrative commands.

Managing the Server with a GUI or CLI (slide 2)

- Windows

The screenshot displays the Windows Server Manager interface for a local server named WIN-MSITE54SFL9. The left-hand navigation pane includes links to the Dashboard, Local Server (which is currently selected), All Servers, and File and Storage Spaces. The main content area is divided into two sections: PROPERTIES and EVENTS.

PROPERTIES
For WIN-MSITE54SFL9

Property	Value	Property	Value
Computer name	WIN-MSITE54SFL9	Last installed updates	8/6/2020 10:03
Workgroup	WORKGROUP	Windows Update	Never check for updates
		Last checked for updates	8/6/2020 12:01
Windows Defender Firewall	Public: Off	Windows Defender Antivirus	Real-Time Protection: On
Remote management	Enabled	Feedback & Diagnostics	Settings: Off
Remote Desktop	Disabled	IE Enhanced Security Configuration	Off
NIC Teaming	Disabled	Time zone	(UTC-08:00) Pacific Time (US & Canada)
Ethernet	192.168.1.100, IPv6 enabled	Product ID	Not activated
Ethernet 2	IPv4 address assigned by DHCP, IPv6 enabled		
Operating system version	Microsoft Windows Server 2019 Standard	Processors	Intel(R) Xeon(R) CPU E-2278 @ 3.40GHz
Hardware information	Microsoft Corporation Virtual Machine	Installed memory (RAM)	4 GB
		Total disk space	110 GB

EVENTS
All events | 14 total

Filter: [Search] [Icons]

Server Name	ID	Severity	Source	Log	Date and Time
WIN-MSITE54SFL9	108	Warning	MSISCSI	System	2/3/2021 6:08:08 AM
WIN-MSITE54SFL9	108	Warning	MSISCSI	System	2/3/2021 6:05:48 AM
WIN-MSITE54SFL9	121	Warning	MSISCSI	System	2/3/2021 6:01:02 AM

Windows Server Manager. (Screenshot courtesy of Microsoft.)

Managing the Server with a GUI or CLI (slide 3)

- Remote administration
 - SSH
 - RDP
 - VNC

Guidelines for Installing an Operating System

- Hardware
- Automate
- Image deployments
- VM templates
- Cloud deployments

Review Activity: OS Installation



Review Activity

Topic 7B

Configure Storage

Exam Objectives Covered

- 2.1 Given a scenario, install server operating systems

Volumes and Partitions

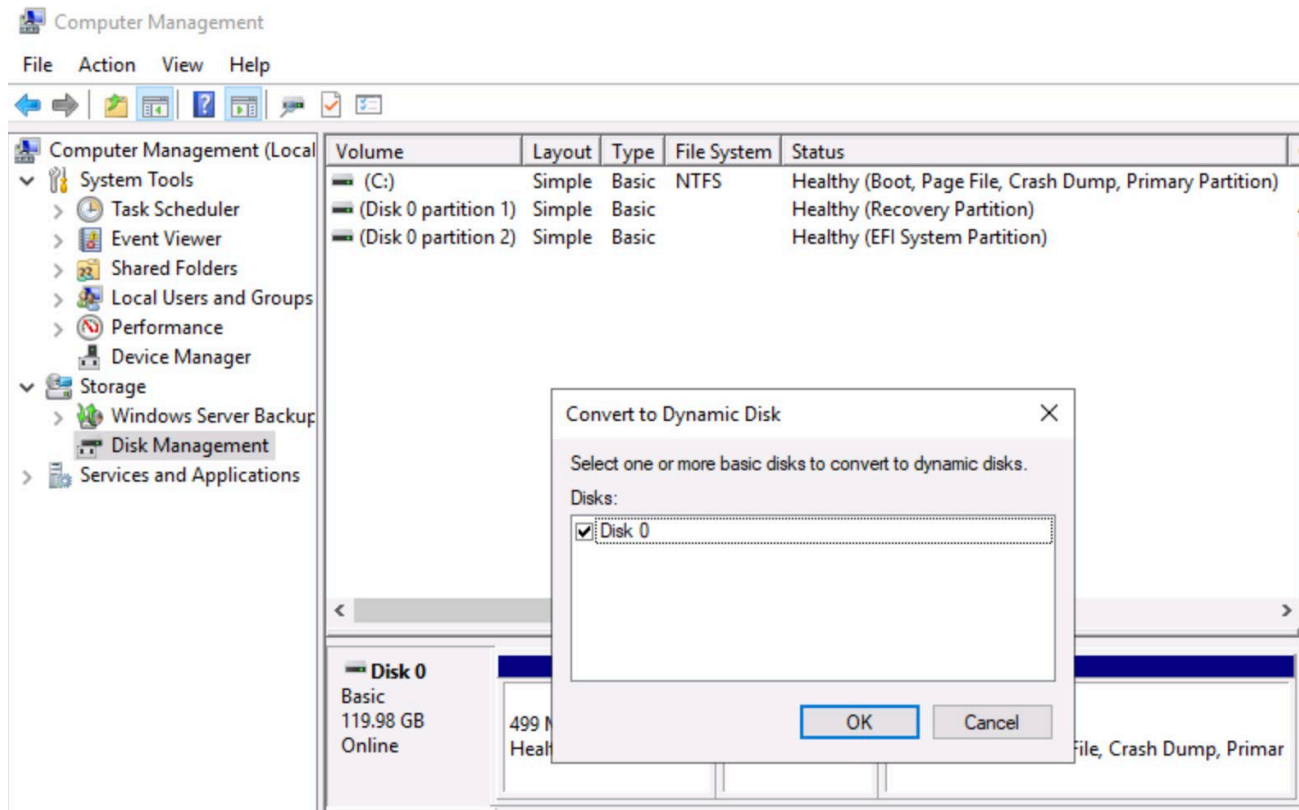
- Windows
- Linux

Partition Tables

- Master Boot Record (MBR)
- Guid Partition Table (GPT)

Partition and Volume Management

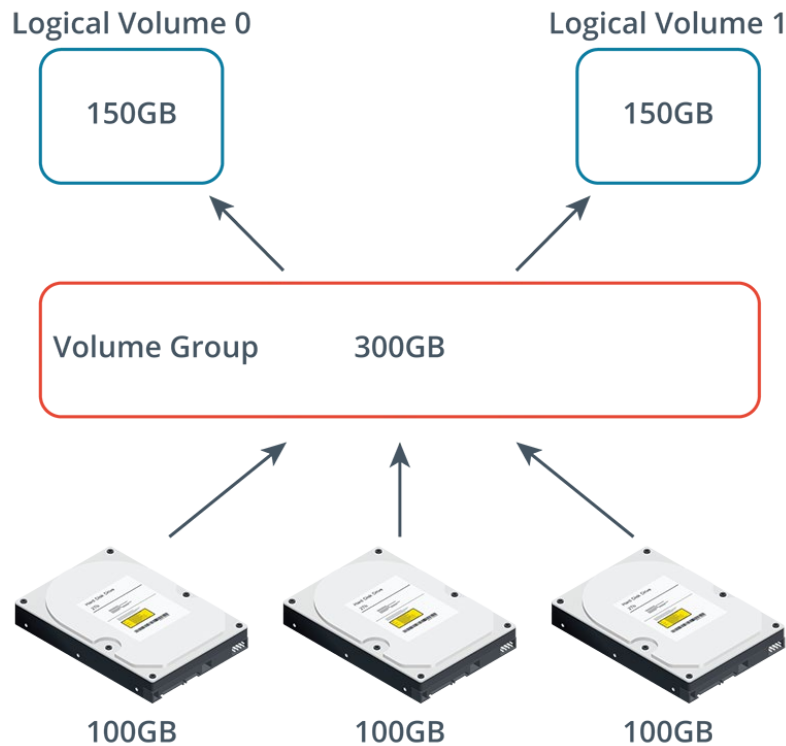
- Partitions
 - Primary
 - Extended
- Windows disks
 - Basic
 - Dynamic



The conversion option in the Windows Server Disk Management console. (Screenshot courtesy of Microsoft.)

Partition and Volume Management (slide 2)

- LVM in Linux
- PVs
- VGs
- LVs



HDDs configured as Physical Volumes

The three layers of LVM. (Images © 123RF.com)

Partition Strategies

- Windows
- Linux
- Windows File Server
- Linux Web Server

File Systems

- ext4
- XFS

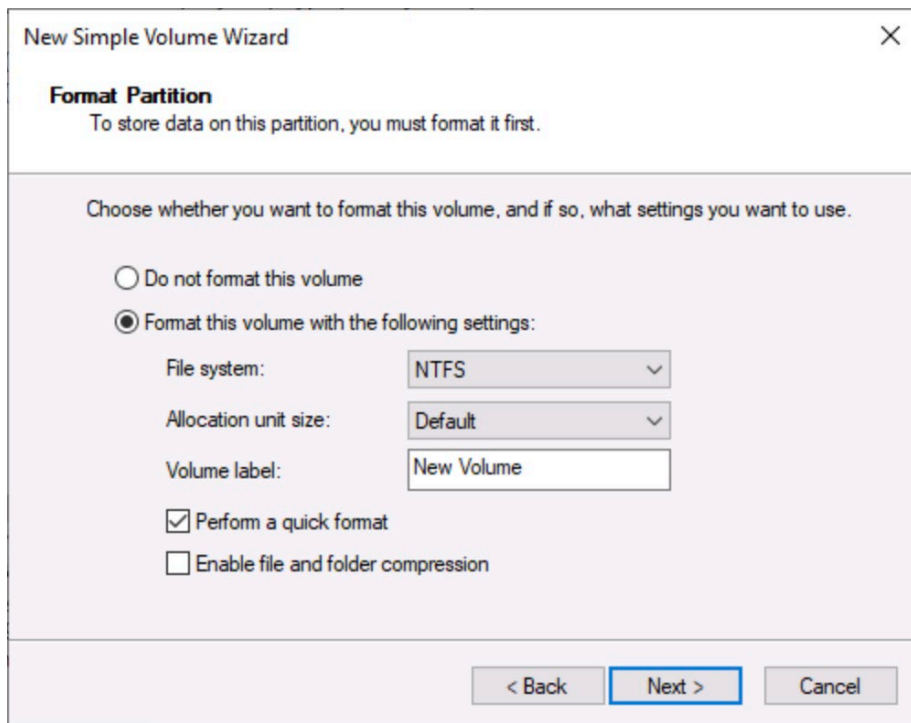
```
[root@CentOS7-A ~]# xfs_info /dev/sda3
meta-data=/dev/sda3            isize=512    agcount=4, agsize=1310720 blks
                         =      sectsz=4096   attr=2, projid32bit=1
                         =      crc=1        finobt=0 spinodes=0
data      =                    bsize=4096   blocks=5242880, imaxpct=25
                         =      sunit=0      swidth=0 blks
naming    =version 2           bsize=4096   ascii-ci=0 ftype=1
log       =internal            bsize=4096   blocks=2560, version=2
                         =      sectsz=4096   sunit=1 blks, lazy-count=1
realtime  =none                extsz=4096   blocks=0, rtextents=0
[root@CentOS7-A ~]# █
```

XFS configuration information on a Linux server.

- ZFS

File Systems (slide 2)

- NTFS
- ReFS
- VMFS



The screenshot shows the 'New Simple Volume Wizard' window, specifically the 'Format Partition' step. The window title is 'New Simple Volume Wizard' with a close button (X) in the top right corner. Below the title bar, the text 'Format Partition' is displayed, followed by the instruction 'To store data on this partition, you must format it first.' The main area of the wizard contains the text 'Choose whether you want to format this volume, and if so, what settings you want to use.' There are two radio button options: 'Do not format this volume' (which is unselected) and 'Format this volume with the following settings:' (which is selected). Under the selected option, there are three settings: 'File system:' set to 'NTFS' (via a dropdown menu), 'Allocation unit size:' set to 'Default' (via a dropdown menu), and 'Volume label:' set to 'New Volume' (via a text input field). Below these settings, there are two checkboxes: 'Perform a quick format' (which is checked) and 'Enable file and folder compression' (which is unchecked). At the bottom of the window, there are three buttons: '< Back' (disabled), 'Next >' (highlighted with a blue border), and 'Cancel' (disabled).

Format a new volume with the NTFS file system. (Screenshot courtesy of Microsoft.)

Guidelines for Configuring Storage

- GPT for > 2TB
- Windows Server has basic and dynamic disk options and uses NTFS
- Linux leverages LVM and uses XFS and ext4
- Windows OS on C:\ and user data on D:\
- Linux uses root (/), home, and var

Review Activity: Storage Configuration



Review Activity

Topic 7C

Configure Network Settings

Exam Objectives Covered

- 2.2 Given a scenario, configure servers to use network infrastructure services

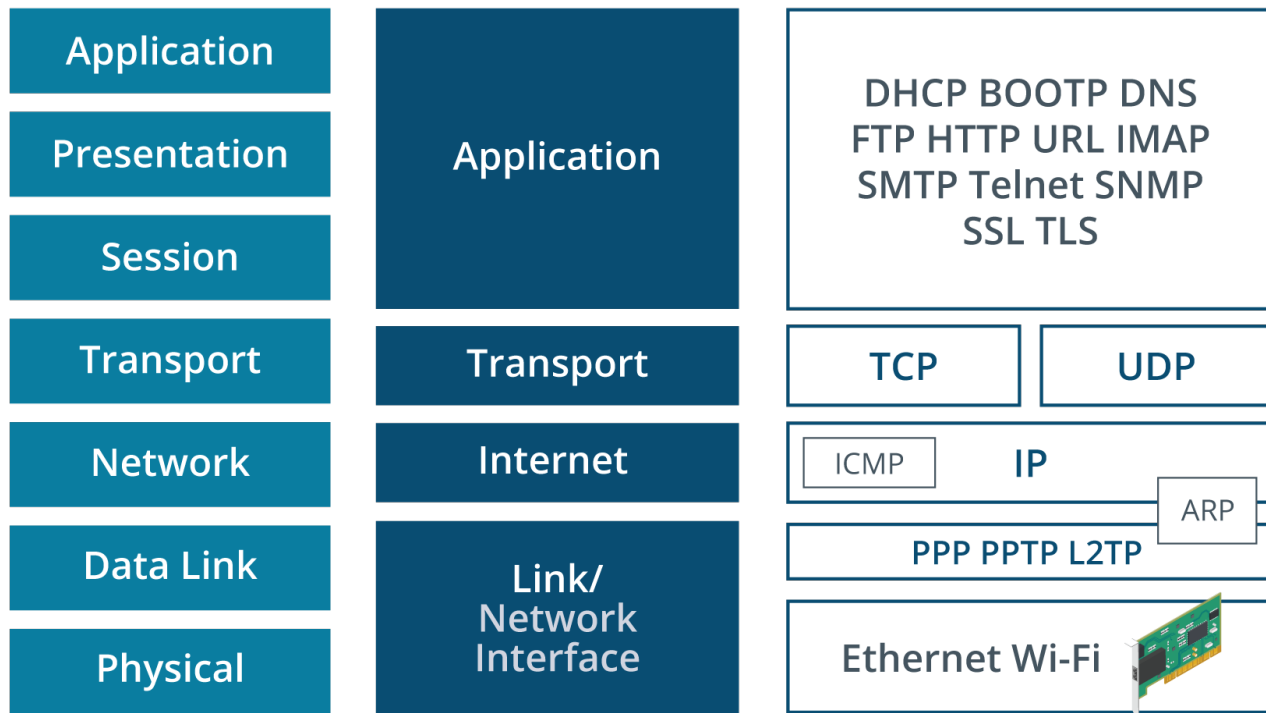
Review Network Fundamentals

- OSI model
- TCP/IP suite

Review the OSI Model

- Application
- Presentation
- Session
- Transport
- Network
- Data Link
- Physical

Review the TCP/IP Suite



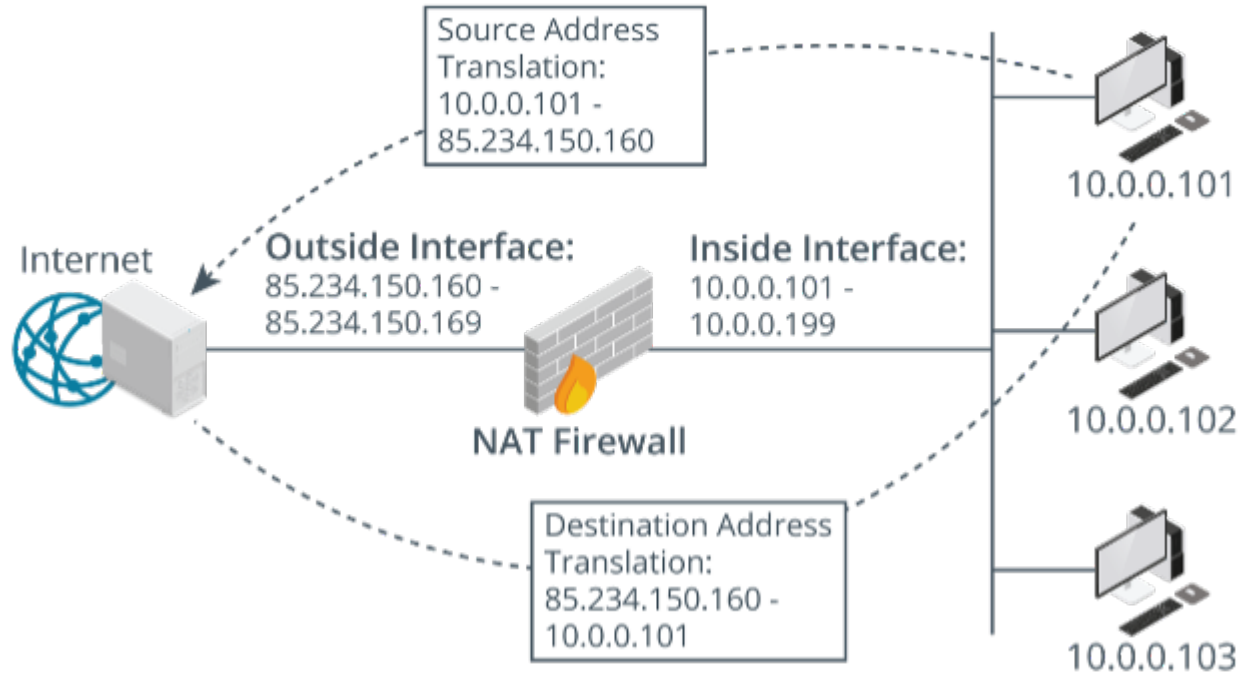
The Application, Transport, Internet, and Network Access Layers of TCP/IP. (Images © 123RF.com)

Addressing

- IPv4 addresses
- The loopback and APIPA addresses
- Network ID, host ID, and the subnet mask
- Bit count notation

Addressing (slide 2)

- Network address translation



NAT device translating between public and private IP addresses. (Images © 123RF.com)

Addressing (slide 3)

- IPv4 address depletion
- IP address management
- IPv6 addresses

```
PS C:\Users\Administrator> ipconfig
```

```
Windows IP Configuration
```

```
Ethernet adapter Ethernet:
```

```
Connection-specific DNS Suffix  . : localdomain
Link-local IPv6 Address . . . . . : fe80::217f:3a98:d74b:ca21%13
IPv4 Address. . . . . : 192.168.1.100
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::1:1%13
                             192.168.1.1
```

Output from the ipconfig command displaying the IPv6 address.

Network Identities

- MAC addresses
- IP addresses
- Hostnames

```
PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

Host Name . . . . . : WIN-MSITE54SFL9
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : localdomain


Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : localdomain
Description . . . . . : Microsoft Hyper-V Network Adapter
Physical Address. . . . . : 00-15-5D-01-80-08
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::217f:3a98:d74b:ca21%13(Preferred)
IPv4 Address. . . . . : 192.168.1.100(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::1:1%13
                             192.168.1.1
DHCPv6 IAID . . . . . : 100668765
DHCPv6 Client DUID. . . . . : 00-01-00-01-25-B2-71-FD-00-15-5D-01-80-08
DNS Servers . . . . . : 8.8.8.8
                             4.4.4.4
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
                             localdomain
```

Ipconfig /all output, note the physical (MAC) address, the IPv4 Address, and the Host Name.

IP Address Configuration

- IP address configuration methods
- Static method

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 100

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 8 . 8 . 8 . 8

Alternate DNS server: 4 . 4 . 4 . 4

☐ Validate settings upon exit

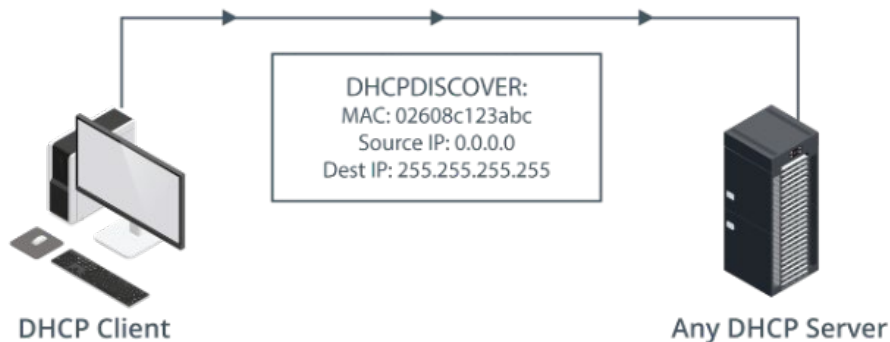
Advanced...

OK Cancel

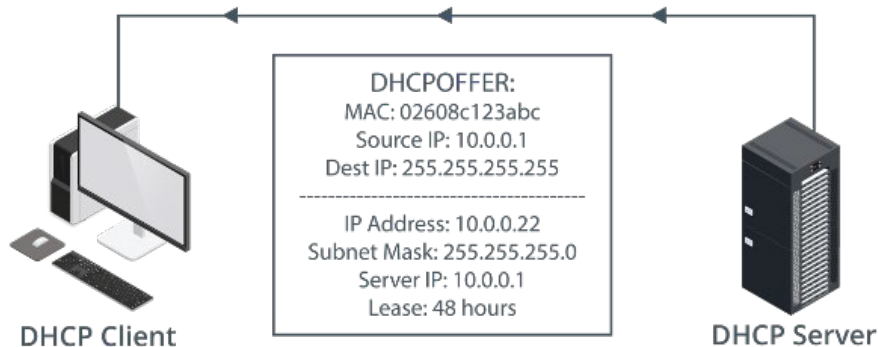
The configuration dialog box to statically set an IP address in Windows. (Screenshot courtesy of Microsoft.)

IP Address Configuration (slide 2)

- Dynamic method



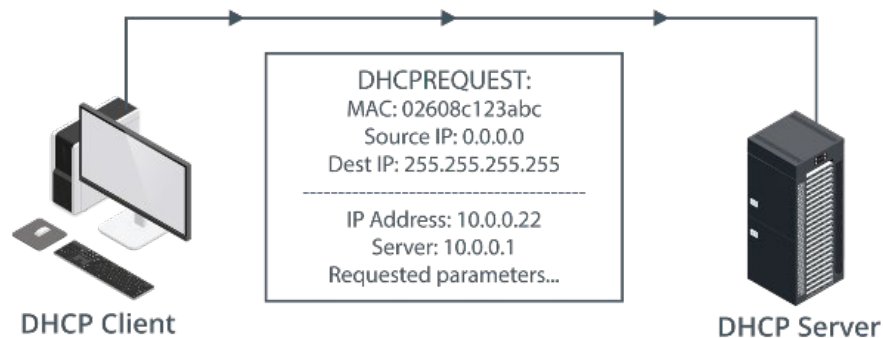
Step one of the four-step DHCP lease generation process: Discover (Images © 123RF.com)



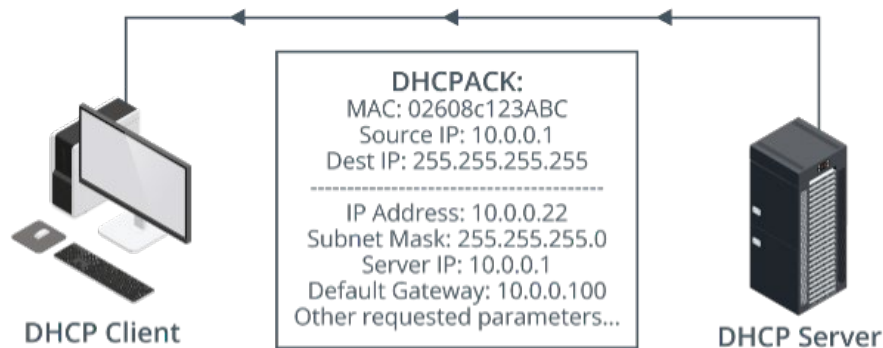
Step two of the four-step DHCP lease generation process: Offer (Images © 123RF.com)

IP Address Configuration (slide 3)

- Dynamic method



Step three of the four-step DHCP lease generation process: Request (Images © 123RF.com)



Step four of the four-step DHCP lease generation process: Acknowledge (Images © 123RF.com)

IP Address Configuration (slide 4)

- DHCP lease
- APIPA address range (169.254.0.0)

```
C:\Users\LabUser>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-BT4131D
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : localdomain

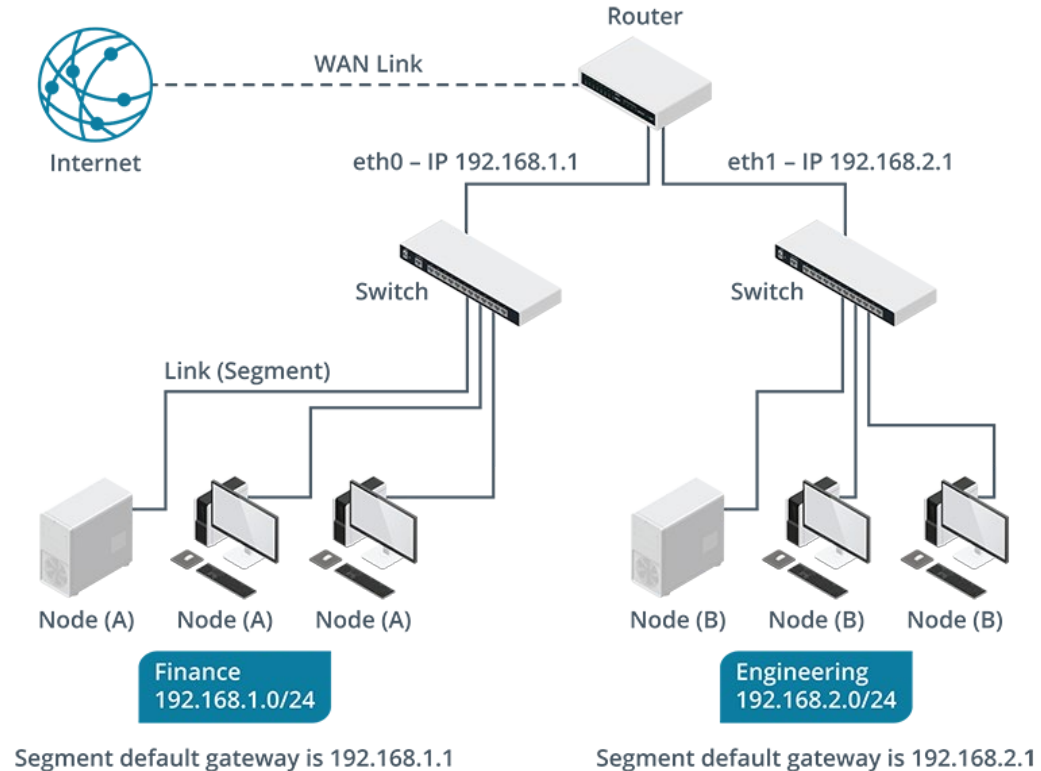

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : localdomain
Description . . . . . : Microsoft Hyper-V Network Adapter
Physical Address. . . . . : 00-15-5D-01-80-00
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::35a2:35c7:3721:90fe%14(Preferred)
IPv4 Address. . . . . : 192.168.1.101(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, February 4, 2021 7:20:31 PM
Lease Expires . . . . . : Thursday, February 4, 2021 9:20:31 PM
Default Gateway . . . . . : fe80::1:1%14
                             192.168.1.1
DHCP Server . . . . . : 192.168.1.1
```

DHCP lease information displayed by using the ipconfig /all command.

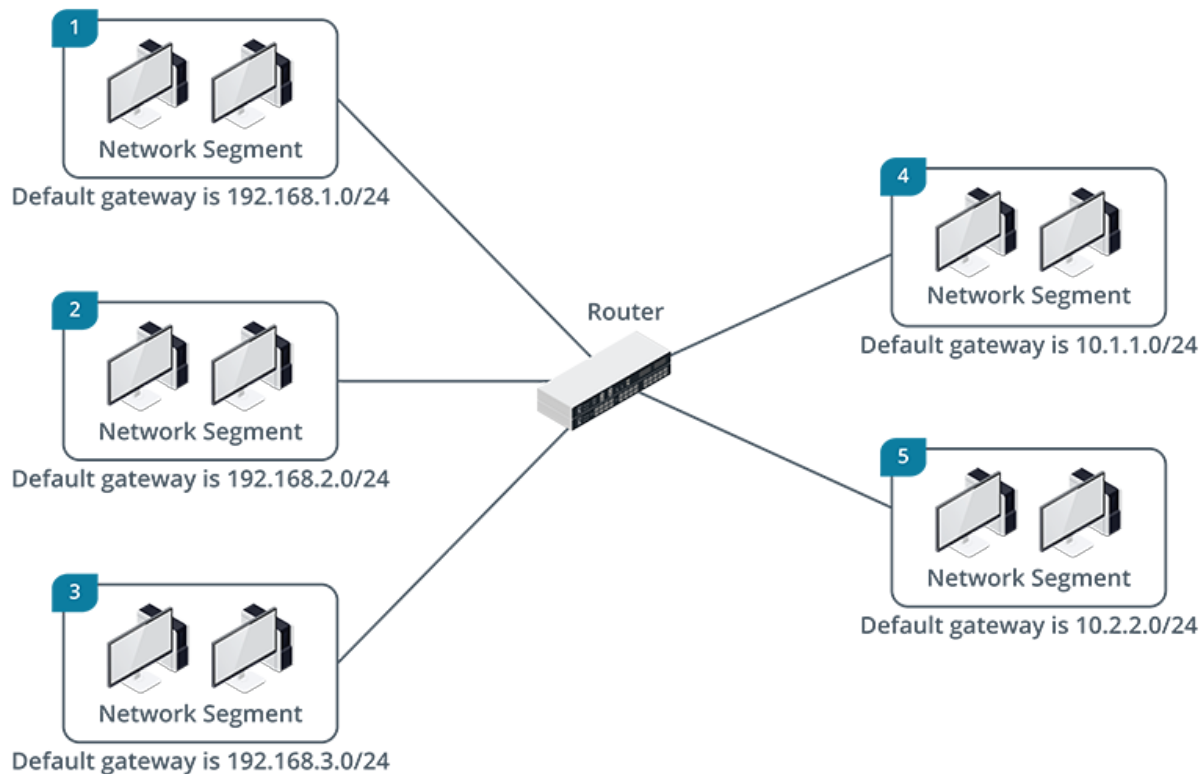
Network Segmentation

- Segmentation



A simple network displaying the router's IP addresses for each NIC, the network IDs and default gateways. (Images © 123RF.com)

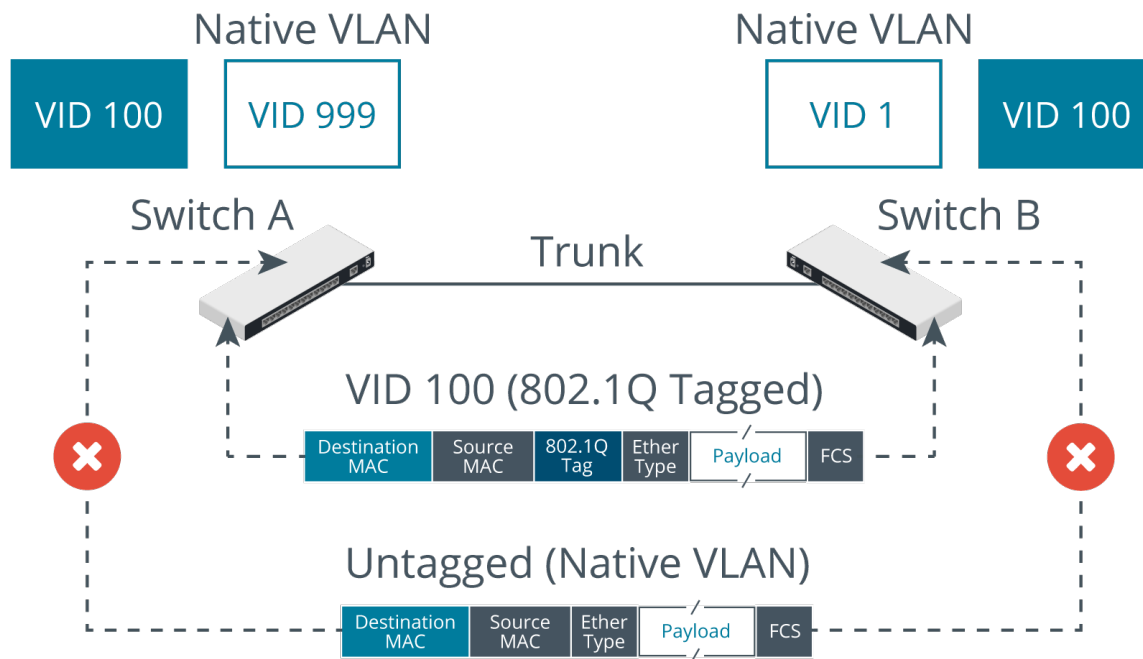
Network Segmentation (slide 2)



A network with five segments and one router. (Images © 123RF.com)

Network Segmentation (slide 3)

- VLAN



A VLAN with multiple switches. (Images © 123RF.com)

Name Resolution

```
[root@localhost ~]# host www.comptia.org
www.comptia.org is an alias for production-comptiawebsite.trafficmanager.net.
production-comptiawebsite.trafficmanager.net is an alias for production-northcentral-www.comptia.org.
production-northcentral-www.comptia.org has address 23.96.239.26
[root@localhost ~]# nslookup 52.165.6.154
** server can't find 154.6.165.52.in-addr.arpa: NXDOMAIN
```

A website URL resolved to an IP address.

```
[root@localhost ~]# █
```

Name Resolution (slide 2)

- FQDN
- Hosts files

```
[root@localhost ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.2.200 workstation01 workstation01.localdomain      #User workstation
192.168.2.201 devworkstation devworkstation.localdomain    #Developer workstation
[root@localhost ~]# █
```

The /etc/hosts file.

Name Resolution (slide 3)

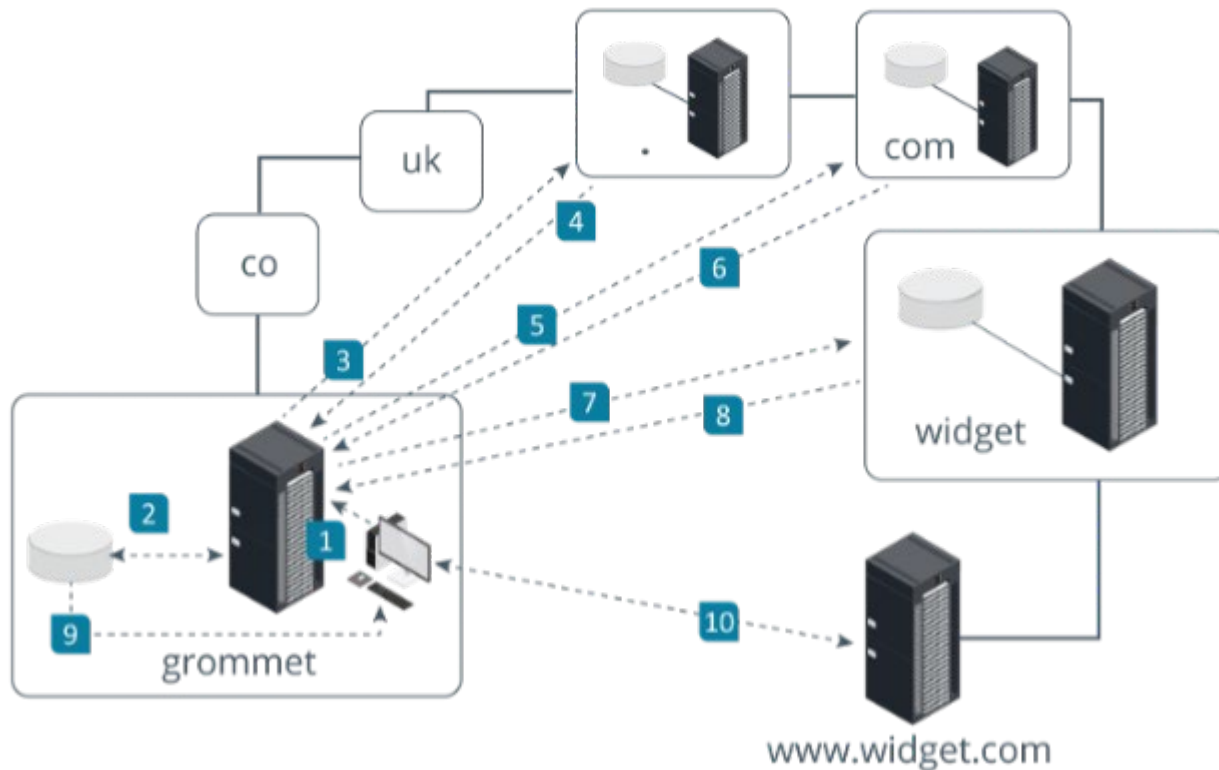
- DNS

 (same as parent folder)	Host (A)	192.168.0.102	30/12/2017
 99servera	Host (A)	192.168.0.188	static
 FileServer1	Host (A)	192.168.0.178	static
 MailServer1	Mail Exchanger (MX)	[10] serverb99.widgets.co...	static
 MailServer2	Mail Exchanger (MX)	[10] servera99.widgets.co...	static
 ServerA99	Host (A)	192.168.0.12	static
 serverb99	Host (A)	192.168.0.102	static
 ServerC99	Host (A)	192.168.0.34	static
 TestServer	IPv6 Host (AAAA)	2001:0001:3dfe:0001:0000:...	static
 www	Alias (CNAME)	serverc99.widgets.com.	static

The DNS records on a Windows Server DNS Server. (Screenshot courtesy of Microsoft.)

Name Resolution (slide 4)

- DNS



Example of a client computer querying DNS to resolve the IP address for a destination server identified by name. (Images © 123RF.com)

Network Devices

- Switch
 - Router
 - WAP
-
- Firewalls and port numbers

Guidelines for Configuring Network Settings

- OSI model
- TCP/IP stack
- Common port numbers
- A node has a MAC address, IP address, and hostname
- Network ID, host ID, subnet mask
- Reserved IP address ranges
- Loopback and IPIPA addresses
- DHCP lease: Discover, Offer, Request, Acknowledge (DORA)
- Default gateway
- VLAN
- Name resolution

Review Activity: Network Settings



Review Activity

Lesson 7 pt. 1

Summary

